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14 December 1954
SUPERSEDING
Army 57-104-2D
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MILITARY SPECIFICATION

FORGINGS FOR COMMON STEEL SHELL

1. SCOPE

1.1 This specification covers forgings for shell (except for shell for smooth-bore weapons) and base plugs made from billets or bars.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications, and the Ordnance Corps drawing pertaining to the forging under contract, and all specifications and drawings referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-M-151 -Metals; General Specification for Inspection of

MILITARY

MIL-G-2550 -General Specification for Ammunition Except Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-The forgings shall be made from material manufactured in accordance with the steel specification specified on the forging drawing, or as specified in the contract or purchase order.

3.2 The forgings shall comply with all requirements on the applicable drawing, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 In order to check the quality of the steel that is to be used in forging blanks, all material shall be examined as follows:

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3.3.1 Fracture (applicable only to blanks divided by nicking and breaking.)

3.3.1.1 Single length blanks.-The base end of each blank after fracture shall expose a clear fresh surface, centrally located, of not less than one-sixth the area of the bar from which the blank is made and shall be free from piping, porosity, and spongy centers which might impair the strength and soundness of the base of the shell when tested as specified in 4.5. The nose end of each blank may be separated by any method.

3.3.1.2 Double length blanks for shell (up to and including 105-MM).-The base ends of each double length blank after fracture shall expose a clear fresh surface, centrally located, of not less than one-sixth the area of the bar from which the blank is made and shall be free from piping, porosity, and spongy centers which might impair the strength and soundness of the base of the shell when tested as specified in 4.5. Ends of blank used for nose of shell may be separated by any method.

3.3.2 Macrostructure (applicable to forging blanks divided by a method other than nicking and breaking and double length blanks for shell larger than 105-MM where the fractured ends are not used as the base of the shell).-The ends of each bar or double length blank when macro-etched shall show no evidence of piping, porosity and spongy centers when tested as specified in 4.6.1.1. The alternative of macro-etching both ends of the first and last bars from each ingot may be exercised provided the forger can supply certification of accurate identification of the first and last bars of each ingot.

3.3.3 Shell 8" and larger.-Provided the contractor is both the forger and machiner of the shell the alternative of macro-etching the base end chucking boss discard subsequent to all lathe operations is permitted in lieu of macro-etching forging blanks or double length blank. The chucking boss discards, when macro-etched, shall show no evidence of piping, porosity and spongy centers when tested as specified in 4.6.1.2.

3.4 Hardness.-The Brinell hardness reading taken on each forging shall not exceed 223 when tested as specified in 4.7. In the case of forging manufacturers who machine their own forgings this hardness requirement shall not apply on forgings that they machine themselves.

3.5 All forgings shall be forged to finished dimensions on the inside unless otherwise specified in the contract.

3.6 Forging practice.

3.6.1 The forging practice used by the contractor shall be such as to produce sound forgings. The satisfactoriness of such forging practice shall be demonstrated to the inspector. Whenever the examination of sectionalized forgings, made on a macro-etched section, indicates unsatisfactory forging practice, necessary steps shall be taken to correct the practice.

3.6.2 The forging temperatures shall be controlled by means of furnace pyrometers or other means satisfactory to the inspector in order to assure uniform correct temperatures.

3.6.3 All forgings shall be subjected to a uniform slow cooling process from the forging temperature down to not over 400°F. Forgings may be cooled at an accelerated rate from 400°F. to room temperature, provided sufficient checks are made with a contact pyrometer or by other suitable means to demonstrate to the satisfaction of the inspector that forgings have reached 400°F. or a lower temperature before being subjected to accelerated cooling. Forgings manufacturers who are also machiners of finished shell may cool the forgings that they machine themselves at any rate which will produce the machinability they desire. Forgers who are teamed with machiners of finished shell may, upon agreement between themselves and such machiners also cool their forgings as prescribed above. The

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cooling rate shall in no case result in defects such as cracks, ovalities of cavity or warping, and this fact must be proved to the satisfaction of the inspector.

3.7 Each forging shall be legibly stamped with the forger's code letter and the lot number of the forgings.

3.8 Workmanship.

3.8.1 Loose scale on the interior and exterior of the forgings shall be removed.

3.8.2 The forging shall be sound and free from injurious scores, pits, seams, laminations, pockets, or other imperfections in the cavity.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.

4.1.1 Production lot.-A production lot shall consist of all forgings made from one heat or melt of steel, and treated by the same method.

4.1.2 Miscellaneous lot.-Unless otherwise agreed upon between machiner and forger, forgings which become separated from the lot to which they belong may be shipped by the forger provided they are properly segregated in the car and identified as to heat number by colored paint or some other system of quick identification. The shipping papers and inspection reports shall be forwarded the machiner properly indicating the heat treatment of each heat involved.

4.2 Sampling.-Number of samples for tests as specified in 4.4, shall be as follows: For lots of 2,000 forgings or less, at least one finished forging of each 1,000 or fraction thereof or for lots more than 2,000 forgings, two finished forgings of each 10,000 or fraction thereof shall be selected by the inspector.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Tension tests.

4.4.1 Procedure.

4.4.1.1 Selection of coupons.-Two test coupons equal in width to three times the wall thickness of the forging shall be cut from each forging selected for test. The test coupons shall be taken longitudinally from diametrically opposite sides of the wall of each of the sample forgings. The end of the coupons shall be located approximately in line with the bottom of the cavity. The coupons shall be above the critical range but not over 1600°F and held at that temperature for approximately one hour and then quenched in oil. The temperature of drawing and other details not covered herein are at the forging manufacturer's option but in no case shall the drawing (tempering) temperature be less than 900°F unless otherwise authorized by the shell producer. (Note.-In making this test, the manufacturer may make any number of preliminary tests for determining the proper treatment of the test coupons to obtain the specified physical properties.)

4.4.1.2 Preparation of test specimens.-One tension test specimen shall be machined from the center of each test coupon. Tension test specimens shall conform to Type 1, Federal Specification QQ-M-151, or, if the wall thickness of the forging is not sufficient for Type 1, to the largest size of Type 4 specimen obtainable.

4.4.1.3 Mechanical properties.-The test coupons shall be tested in a manner satisfactory to the inspector to determine compliance with the requirement specified on the applicable drawing. The yield strength shall be determined by the extension-under-load-method, as specified in Specification QQ-M-151.

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The limiting extension-under-load shall be calculated as follows:

$$X = \frac{Y}{30,000,000} + 0.001$$

where:

X = limiting extension, inch per inch of gage length
Y = specified yield strength

4.4.2 Acceptance.-The lot shall be considered acceptable provided the test specimens representing a lot of forgings comply with the mechanical requirements specified on the applicable drawing. If any test specimen fails to comply with the requirement specified on the drawing, the forging lot shall be rejected.

4.5 Fracture test (applicable only to blanks divided by nicking and breaking).

4.5.1 Procedure.-The bars and billets shall be divided into forging blanks. The contractor's process for dividing the blanks must not develop injuries in the blank which may in turn result in a defective shell. Observation shall be made to determine compliance with 3.3.1.1 or 3.3.1.2 as applicable.

4.5.2 Acceptance.-The blanks shall be considered acceptable provided they comply with the requirement of 3.3.1.1 or 3.3.1.2 as applicable. If any blank fails to comply with this requirement the blank shall be rejected.

4.6 Macrostructure (applicable to forging blanks divided by a method other than nicking and breaking and double length blanks for shell larger than 105-MM where the fractured ends are not used as the base of the shell).

4.6.1 Procedure.

4.6.1.1 Bar or double length blank.-A full cross section shall be taken from both ends of each bar from which blanks are to be made or from both ends of a double length blank or from both ends of the first and last bars from each ingot provided certification is furnished as indicated in 3.3.2. These sections shall measure not less than one-half inch in the direction of the axis of the bar, and may be cut long enough to forge into shell to avoid unnecessary waste of material. The faces representing the cross-section of the bar shall be finished and etched in a manner satisfactory to the inspector. (See 6.2). Observation shall be made of each bar or double length blank to determine compliance with 3.3.2.

4.6.1.2 Chucking boss.-The chucking boss removed from each shell shall be at least one-half inch thick measured in the longitudinal direction of the shell axis and at least one-third the shell diameter with its center corresponding to that of the shell base and shall be sawed within one-eight inch of the finished shell base end. Corresponding identifying marks shall be placed on the chucking boss and its respective shell for the purpose of rejecting that shell in the event of failure of the chucking boss to comply with the requirement of 3.3.3. The cut surface (shell-side surface) of the chucking boss shall be polished and etched in a manner satisfactory to the inspector. (See 6.2). Observation shall be made of each chucking boss to determine compliance with 3.3.3.

4.6.2 Acceptance.

4.6.2.1 Bar and double length blank.-The bars or double length blanks shall be considered acceptable provided each bar of double length blank complies with the requirement of 3.3.2. The entire bar or double length blank shall be rejected if one end shows injurious defects. If the option of macro-etching both ends of the first and last bars from each ingot is exercised (See 3.3.2) and should any of the ends show defects as specified in 3.3.2 all bars from that ingot shall be

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macro-etched on both ends. The contractor may remove the defective portion of the bar or double length blank and resubmit it for re-examination. The re-submitted bars or double length blanks shall be acceptable provided they comply with the requirements of 3.3.2. The shell from which the chucking boss has been removed shall be considered acceptable provided the chucking boss complies with the requirement of 3.3.3. If any chucking boss fails to comply with this requirements, the shell from which the chucking boss has been removed shall be rejected.

4.7 Hardness test.

4.7.1 Procedure.-Two tests, approximately 180° apart, shall be made on the exterior of each forging selected for test, approximately 2 inches from the base end. Tests shall be made on a sufficient percentage of each lot of forging to insure that uniformity is being maintained and that the cooling process results in good forgings. Observation shall be made to determine compliance with 3.4.

4.7.2 Acceptance.-The lot shall be considered acceptable provided the sample forgings comply with the requirement of 3.4. If any forging fails to comply with this requirement the forging lot shall be rejected.

4.8 Forging practice test.-On the first lot of any new contract one forging out of each 100 produced shall be sectionalized for examination (see 3.6.1), except when the new contract immediately follows a previous contract under which satisfactory material of the same character was manufactured. On subsequent lots when, in the opinion of the inspector, the forging practice used by the contractor is producing forgings of questionable quality, he may select for sectionalizing and examination such forgings as he deems necessary to determine the quality of the forging in question. In addition, all forging from which tensile test coupons are cut shall be sectionalized longitudinally to show the middle of the base for examination. For this examination the section shall be macro-etched to the satisfaction of the inspector.

4.9 Reports.

4.9.1 A report of the results of all tests made in connection with the acceptance of each heat of steel, together with the forger's report of tension tests and details of heat treatment employed for each lot, shall be furnished the consignee by the manufacturer with invoice. These reports shall embody the results obtained by the steel maker and the forger for the use of the machiner.

4.9.2 Reports on standard form prescribed by the Ordnance Corps covering chemical composition, tension tests, and details of heat treatment employed for each lot, shall be furnished the inspector.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Separation.-All forgings shall be properly separated by lots when loaded for shipment in order that the inspector at the receiving end may readily identify them.

5.1.2 All shipments shall be made in box cars, gondola cars, trucks, or other approved carrier, unless otherwise directed in writing by the contracting officer. All shipments shall be loaded in such a manner as to insure acceptance, by common or other carrier, for safe transportation at the lowest rate to the point of delivery.

5.1.3 Marking of shipments shall be such as to provide for proper delivery and identification.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) The applicable Ordnance Corps drawing number.
(c) Whether finished dimensions are required on the inside (see 3.5).
(d) Whether miscellaneous lots are permitted (see 4.1.2).

6.2 The suggested acid mixture, etching time and etching temperature for free cutting steels is as follows:

Acid: 38 percent HCL-12 percent H_2SO_4-50 percent H_2O

or

50 percent HCL-50 percent H_2O.
Etching time: Approximately 15 minutes.
Etching temperature: 160°F.-180°F.

For plain carbon steels the foregoing acid mixtures are satisfactory but the etching time and temperature must be such as will produce a satisfactory etch.

6.3 The inspector will be provided with photographs of specimens of unsatisfactory forging practice and macro-structure.

NOTICE.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded, by implication or otherwise, as in any manner licensing the holder, or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
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